



SURFACE VEHICLE STANDARD

J1979™-2

APR2021

Issued

2021-04

E/E Diagnostic Test Modes: OBDOnUDS

RATIONALE

Due to regulatory adoption of OBDOnUDS, SAE J1979 now becomes a multiple part document series:

1. SAE J1979, which defines diagnostic test modes for legacy data links used for emissions OBD (e.g., SAE J1850, ISO 9141-2, ISO 14230-4, ISO 15765-4).
2. SAE J1979-2, which defines the latest diagnostic services for OBDOnUDS used for emissions OBD (e.g., ISO 14229-1).

Market regulations specify which data links are allowed for sale in a given market.

OBDOnUDS is not just a translation of the classic SAE J1979 \$01-\$0A services to UDS services—it provides extra features: DTC based Readiness, ExtendedDataRecords for DTCs, Readiness group to DTC readiness translation, and support for snapshots for up to five DTCs and two occurrences each. Other clarifications and functional enhancements are also included in this document.

FOREWORD

SAE J1979-2 specifies diagnostic services and functionally addressed request/response messages required to be supported by motor vehicles and external test equipment for diagnostic purposes which pertain to motor vehicle emission-related data. These messages are intended to be used by any external test equipment meeting the requirements of SAE J1978 for retrieval of OBD information from a vehicle.

SAE Executive Standards Committee Rules provide that: "This report is published by SAE to advance the state of technical and engineering sciences. The use of this report is entirely voluntary, and its applicability and suitability for any particular use, including any patent infringement arising therefrom, is the sole responsibility of the user."

SAE reviews each technical report at least every five years at which time it may be revised, reaffirmed, stabilized, or cancelled. SAE invites your written comments and suggestions.

Copyright © 2021 SAE International

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without the prior written permission of SAE.

TO PLACE A DOCUMENT ORDER: Tel: 877-606-7323 (inside USA and Canada)
Tel: +1 724-776-4970 (outside USA)
Fax: 724-776-0790
Email: CustomerService@sae.org
http://www.sae.org

SAE WEB ADDRESS:

For more information on this standard, visit
https://www.sae.org/standards/content/J1979-2_202104

TABLE OF CONTENTS

1.	SCOPE.....	8
1.1	Introduction	8
1.1.1	SAE Document Reference Concept	9
1.1.2	SAE J1979DA (Digital Annex)	11
1.1.3	SAE Digital Annex Revision Procedure	11
1.2	Document Scope.....	12
2.	REFERENCES.....	12
2.1	Applicable Documents	12
2.1.1	SAE Publications.....	12
2.1.2	ISO Publications.....	13
3.	TERMS, DEFINITIONS, SYMBOLS, AND ABBREVIATED TERMS	13
4.	CONVENTIONS	18
5.	DOCUMENT OVERVIEW	19
6.	TECHNICAL REQUIREMENTS.....	20
6.1	General Requirements	20
6.1.1	OBDOnUDS Enhanced Information	21
6.2	Diagnostic Service Requirements	24
6.2.1	Protocol Detection.....	24
6.2.2	Multiple Responses to a Single Data Request	25
6.2.3	Application Timing Parameter Definition.....	26
6.2.4	Minimum Time Between Requests from External Test Equipment	27
6.2.5	Data Not Available	28
6.2.6	Maximum Values.....	31
6.2.7	Invalid Signals	32
6.3	Diagnostic Message Format	32
6.3.1	Maximum Message Length.....	32
6.3.2	Response Code Parameter Definition	32
6.3.3	Header Byte Definition of ISO 15765-4.....	33
6.3.4	Data Bytes Definition of ISO 15765-4.....	33
6.4	Byte Order Convention.....	33
6.5	Allowance for Expansion and Enhanced Diagnostic Services.....	33
6.6	Definition of PIDs for Service \$22 ReadDataByIdentifier and \$19 \$04 ReadDTCInformation	34
6.7	Format of Data to be Displayed	34
7.	DIAGNOSTIC SERVICE DEFINITION FOR OBDONUDS.....	35
7.1	Request Current Powertrain Diagnostic Data Using Service \$22	35
7.1.1	Functional Description.....	35
7.1.2	Message Data Bytes	35
7.1.3	Parameter Definition	38
7.1.4	Message Example.....	38
7.2	Request Powertrain Freeze Frame Data Using Service \$19 Subfunction \$04.....	43
7.2.1	Functional Description.....	43
7.2.2	Message Data Bytes	45
7.2.3	Parameter Definition	46
7.2.4	Message Example.....	47
7.3	Request Emission-Related Diagnostic Trouble Codes with Confirmed Status Using Service \$19 - Subfunction \$42.....	48
7.3.1	Functional Description.....	48
7.3.2	Message Data Bytes	49
7.3.3	Parameter Definition	50
7.3.4	Message Example.....	51

7.4	Clear/Reset Emission-Related Diagnostic Information Using Service \$14	54
7.4.1	Functional Description.....	54
7.4.2	Message Data Bytes	55
7.4.3	Parameter Definition	55
7.4.4	Message Example.....	56
7.5	Request Oxygen Sensor Monitoring Test Result Service - Request Oxygen Sensor Monitoring Test Result (Not Supported)	56
7.6	Request On-Board Monitoring Test Results for Specific Monitored Systems Using Service \$22 - MID	56
7.6.1	Functional Description.....	56
7.6.2	Message Data Bytes	58
7.6.3	Parameter Definition	60
7.6.4	Message Example.....	65
7.7	Request Emission-Related Diagnostic Trouble Codes with Pending Status Using Service \$19 - Subfunction \$42	67
7.7.1	Functional Description.....	67
7.7.2	Message Data Bytes	68
7.7.3	Parameter Definition	69
7.7.4	Message Example.....	69
7.8	Request Control of On-Board System, Test, or Component Using Service \$31	70
7.8.1	Functional Description.....	70
7.8.2	Message Data Bytes	70
7.8.3	Parameter Definition	72
7.8.4	Message Example.....	72
7.9	Request Vehicle Information using Service \$22 - DID.....	76
7.9.1	Functional Description.....	76
7.9.2	Message Data Bytes	76
7.9.3	Parameter Definition	78
7.9.4	Message Example.....	78
7.10	Request Emission-Related Diagnostic Trouble Codes with Permanent Status Using Service \$19 - Subfunction \$55	86
7.10.1	Functional Description.....	86
7.10.2	Message Data Bytes	87
7.10.3	Parameter Definitions.....	87
7.10.4	Message Example.....	88
7.11	Request Supported DTCEXtendedRecord Information Using Service \$19 - Subfunction \$1A	88
7.11.1	Functional Description.....	88
7.11.2	Message Data Bytes	88
7.12	Request DTCEXtendedDataRecord Using Service \$19 - Subfunction \$06	88
7.12.1	Functional Description.....	88
7.12.2	Message Data Bytes	89
7.12.3	Message Example.....	89
7.13	Request DTCs for a ReadinessGroup Service \$19 - Subfunction \$56.....	91
7.13.1	Functional Description.....	91
7.13.2	Message Data Bytes	91
7.13.3	Message Example.....	92
8.	ADDRESSING SCHEME	94
8.1	Functional Addressing.....	94
8.1.1	Protocol Identification.....	94
8.1.2	Identify the Vehicle	94
8.1.3	ClearDTCs	94
8.1.4	I/M Inspection.....	94
8.2	Physical Addressing	94
8.3	Use Cases for Physical Addressing	94
8.3.1	Identify the Supported Functionality of the ECU	94
8.3.2	Read Data Stream Parameters.....	94
8.3.3	Read ECU Information or OBD Monitor TestIds	94
8.3.4	Request DTC Information	94
8.3.5	Request DTC freeze Frame Information	95